

Monash University, Faculty of Information Technology

FIT3080

Applied Class for Bayesian Networks I

Please note: Use a calculator or spreadsheet where helpful.

Exercise 1: Joint probabilities, conditional probabilities and Bayes' theorem

When driving into the city, you've noticed a pattern that there is often heavy traffic when there is a major event or it is peak hour. You estimate the joint probabilities as:

MajorEvent (M)	PeakHour (H)	Traffic (T)	P()
Yes	Yes	Heavy	0.005
Yes	Yes	Moderate	0.001
Yes	Yes	Light	0.000
Yes	No	Heavy	0.010
Yes	No	Moderate	0.020
Yes	No	Light	0.004
No	Yes	Heavy	0.085
No	Yes	Moderate	0.045
No	Yes	Light	0.014
No	No	Heavy	0.080
No	No	Moderate	0.240
No	No	Light	0.496

(All answers to 3 decimal places.)

- What is the probability that traffic will be heavy $P(T=\text{Heavy})$?
- What is the probability that traffic will be moderate or heavy $P(T=\text{Heavy} \vee T=\text{Moderate})$?
- What is the probability that there is a major event and it is peak hour $P(M=\text{Yes} \wedge H=\text{Yes})$?
- What is the probability that there is a major event if traffic is heavy $P(M=\text{Yes} | T=\text{Heavy})$?
- Is the level of traffic dependent on a major event?
- Show the conditional distribution for major events and traffic, given it is peak hour $P(M \wedge T | H=\text{Yes})$.
- (Optional)** Are major events dependent on whether it is peak hour?

Exercise 2: Conditional probabilities and Bayes' theorem (Medical condition)

Approximately 1 in 20 people in Australia have type 2 diabetes. An oral glucose tolerance test (OGTT) is a way of testing for the presence of type 2 diabetes and is considered quite accurate. Numerical estimates of its accuracy vary, but are generally close to a true positive rate of 85% and a false positive rate of 5%.

Apply Bayes' theorem to answer the following questions:

- a) If someone tests positive on the OGTT then what is the probability that they have diabetes?
- b) If someone tests negative on the OGTT then what is the probability that they have diabetes?
- c) Given the results, should we rely on the OGTT alone in diagnosing diabetes? In more realistic situations, why might our prior be different to the general prevalence of diabetes?
- d) How would a false positive rate of 1% change our estimate of the probability that someone has diabetes if they test positive?
- e) **(Optional)** Is the test more sensitive to changes in the true positive rate or the false positive rate?

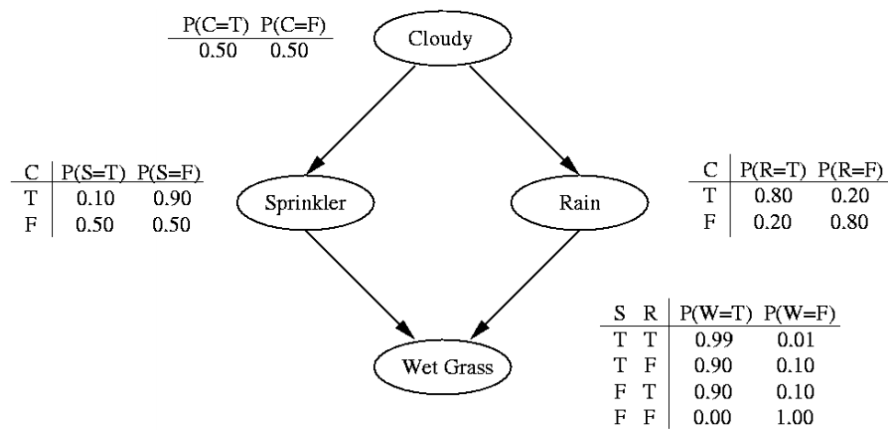
Exercise 3: Bayesian Network (Late to Campus)

Let's assume that you don't have a car, but rather you come to campus either by bike or by bus. If you take the bus, there is a 20% chance that you will be late. If you take the bike, there is a 5% chance that you will be late. You come to campus by bike 4 days out of 5. Today you were late to campus.

- a) Write down and explain the formula used in Bayes' rule for this problem:
 $P(\text{bus}|\text{late})$
- b) Use Bayes' rule to calculate the probability that you took the bus today.
- c) Model the situation as a Bayesian network with 2 nodes and give the conditional probability distribution for both nodes.
- d) Add a third node for setting an alarm. You set an alarm 40% of the time, and it reduces your chances of being late to 1% if travelling by bike and 19% if travelling by bus.
 - a. What is $P(\text{bus}|\text{late}, \text{alarm}=\text{False})$?
 - b. Without calculating, will $P(\text{bus}|\text{late}, \text{alarm}=\text{True})$ be higher, lower or the same as $P(\text{bus}|\text{late}, \text{alarm}=\text{False})$?

Exercise 4: Bayesian Network (Sprinkler)

Consider the following Bayesian network:



The event that the grass being wet ($W=true$) has 2 possible causes. It is either the water sprinkler was on ($S=true$) or it rained ($R=true$).

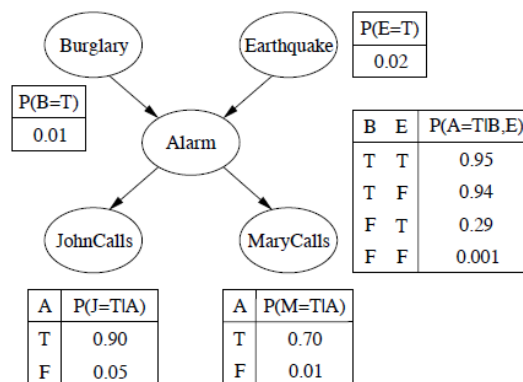
Construct this BN using Netica software. Use it to help you answer the following queries:

- Given the observation that the grass is wet, what is the probability that it is caused by rain?
- Given the observation that the grass is wet, how likely is it due to the water sprinkler was on?

Exercise 5: Bayesian Network (Pearl's Earthquake example) (Netica)

Example statement: You have a new burglar alarm installed. It reliably detects burglary, but also responds to minor earthquakes. Two neighbors, John and Mary, promise to call the police when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the alarm with the phone ringing and calls then also. On the other hand, Mary likes loud music and sometimes doesn't hear the alarm. Given evidence about who has and hasn't called, you'd like to estimate the probability of a burglary (from Pearl (1988)).

A BN representation of this example is shown in Figure 2.6. All the nodes in this BN are Boolean, representing the true/false alternatives for the corresponding propositions. This BN models the assumptions that John and Mary do not perceive a burglary directly and they do not feel minor earthquakes. There is no explicit representation of loud music preventing Mary from hearing the alarm, nor of John's confusion of alarms and telephones; this information is summarized in the probabilities in the arcs from *Alarm* to *JohnCalls* and *MaryCalls*.



[Above taken from Korb&Nicholson, Bayesian Artificial Intelligence, Section 2.5.1]

Construct an Earthquake BN using Netica software. Use it to help you answer the following questions about specific scenarios:

- Only Earthquake is set to True: which nodes are not affected and why?
- Only Alarm is set to True: which nodes are not affected and why?
- With Alarm set to True, Earthquake is set to False:
 - which nodes are affected and why?
 - what is the relationship between Earthquake and Burglary?
 - which nodes are not affected and why?
- Answer these three questions with Alarm set to False.
- With Alarm set to True, John Calls is set to False: which nodes are affected and why?